

1. Input from User (Scanner class)

- Java uses the **Scanner class** (from java.util package) to take input.
- It allows reading different data types like int, double, String, etc.
- Common methods:
 - nextInt() → integer input
 - nextDouble() → decimal input
 - nextLine() → string input

```
Scanner sc = new Scanner(System.in);  
int num = sc.nextInt();
```

2. Conditional Statements

These allow programs to make **decisions**.

(a) if

- Executes a block of code **only if the condition is true**.

Syntax:

```
if (condition) {  
    // statements  
}
```

(b) if-else

- Executes one block if the condition is true, otherwise another block.

Syntax:

```
if (condition) {  
    // true block  
} else {  
    // false block  
}
```

(c) if-else-if ladder

- Used when there are **multiple conditions** to test.
- Conditions are checked one by one.

Syntax:

```
if (condition1) {  
    // block1  
} else if (condition2) {  
    // block2  
} else {  
    // default block  
}
```

(d) Nested if

- An if inside another if.
- Useful when **one condition depends on another**.

Syntax:

```
if (condition1) {  
    if (condition2) {  
        // nested block  
    }  
}
```

(e) switch-case

- A cleaner way to check multiple conditions (works with int, char, String).
- break is used to stop execution after a case.
- default executes if no case matches.

Syntax:

```
switch (variable) {  
    case value1:  
        // block1  
        break;  
    case value2:  
        // block2  
        break;  
    default:  
        // block if no match  
}
```

3. Loops

Loops allow **repeating code** multiple times.

(a) for loop

- Used when we know the exact number of repetitions.

Syntax:

```
for (initialization; condition; increment/decrement) {  
    // statements  
}
```

(b) while loop

- Condition is checked **before** execution.
- Runs until the condition is false.

Syntax:

```
while (condition) {  
    // statements  
}
```

(c) do-while loop

- Executes at least **once**, then checks condition.

Syntax:

```
do {  
    | // statements  
} while (condition);
```

(d) Nested Loops

- A loop inside another loop.
- Used for tables, patterns, grids, etc.

Syntax:

```
for (...) {  
    for (...) {  
        // nested block  
    }  
}
```

4. Break & Continue

(a) break

- Exits from the loop or switch **immediately**.

Syntax:

```
// break;
```

(b) continue

- Skips the **current iteration** and moves to the next one.

Syntax:

```
// continue
```

Real-Time Scenario Based Problems

1. ATM Machine Simulation (Decision + Loops + Break)

- User logs in with a PIN.
- Display menu: Check Balance, Withdraw, Deposit, Exit.
- Use switch-case for menu.
- Use while loop to keep showing menu until user exits.
- Use break to stop when user chooses Exit.

Concepts used: Scanner (input), switch-case, while loop, break.

2. Student Grade Calculator (if-else-if ladder + Nested if)

- Take marks as input.
- If marks $\geq 90 \rightarrow$ Grade A
- Else if $\geq 75 \rightarrow$ Grade B
- Else if $\geq 50 \rightarrow$ Grade C
- Else $\rightarrow$ Fail
- Nested if: If Grade A $\rightarrow$ further check if marks $\geq 95 \rightarrow$ Award "Scholarship Eligible".

Concepts used: if-else-if, nested if, Scanner input.

3. Restaurant Ordering System (switch + loops + continue)

- Show food menu with item numbers.
- User selects item → add price to bill.
- Use switch-case to decide price.
- Keep asking until user enters 0 (Exit).
- If user enters invalid option, use continue to skip and re-show menu.

Concepts used: Scanner input, switch-case, do-while loop, continue.

4. Multiplication Table Generator (for loop + nested loop)

- Take a number from user.
- Print multiplication table using for loop (1 to 10).
- Extend: For multiple tables (e.g., 2 to 5), use **nested loops**.

Concepts used: for loop, nested for loop, Scanner input.

5. Login Authentication System (while loop + if-else + break)

- User must enter correct username & password.
- Allow maximum 3 attempts.
- If correct → show “Login Successful”.
- If wrong after 3 tries → block user.
- Use break once correct login is entered.

Concepts used: while loop, if-else, break.